

directly at the expense of the users, grew from hundreds of thousands of dollars when front-running was first publicly demonstrated in 2017,¹⁶ to hundreds of millions of dollars as of mid-2021.¹⁷ Large transactions, especially in illiquid markets with high slippage, are particularly susceptible to front-running. For this reason, Uniswap allows users to set a maximum slippage as a clause in the transaction. If the acceptable level of slippage is exceeded, the trade will fail to execute.¹⁸ This provides a limit to the profit front-runners can make but does not completely remove the problem.

Another drawback is that arbitrage profits go only to arbitrageurs – they do not have a vested interest in the platform. The arbitrageurs profit at the expense of liquidity providers, who should not be losing the potential spread they would earn in a normal market-making scenario. Competing platforms, such as Mooniswap,¹⁹ propose to solve this issue by supplying virtual prices that slowly approach the true price, leaving tighter time windows and lower spreads for arbitrageurs to capitalize on. The additional spread remains in the pool for the liquidity providers.

Uniswap offers an interesting feature called a *flash swap*, which is similar to a flash loan. In a flash swap, the contract sends the tokens *before* the user pays for them with assets on the other side of the pair, unlocking many opportunities for arbitrageurs. The user can deploy this instant liquidity to acquire the other asset at a discount on another exchange before repaying it; the corresponding amount of the alternate asset must be repaid to maintain the invariant. This flexibility in a flash swap is different from the provision